BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BAHAWALPUR

INTERMEDIATE (11TH CLASS) EXAMINATION, 2024

BIOLOGY

Past Paper 2024 • Paper Code: 6463

Retyped faithfully from the original board paper — same questions, no changes.

Paper I (Objective Type) — Inter (Ist – A – Exam – 2024)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6463

Note: Four choices A, B, C, D to each question are given. Which choice is correct fill that circle in front of that Question No. on the Objective Bubble Sheet. Use Marker or Pen to fill the circles. Cutting or filling two or more circles will result in Zero Mark in that Question.

Q1. Which of the following is true about Birds :

(A) They have right aortic arch
(B) They have left aortic arch
(C) They have right and left aortic arch
(D) They do not have aortic arch

Q2. Hydathodes in plants are associated with :

(A) Transpiration
(B) Guttation
(C) Conduction
(D) Imbibition

Q3. Respiratory Pigment present in Muscles is called :

(A) Myoglobin
(B) Globin
(C) Haemoglobin
(D) Haemocyanin

Q4. In root nodules of Leguminous plants, bacteria convert Nitrogen into :

(A) Ammonia
(B) Nitrate
(C) Urea
(D) Nitrite

Q5. The removal of terminal Phosphate of ATP during hydrolysis releases about ____ of energy :

(A) 7.3 K cal
(B) 6.3 K cal
(C) 5.3 K cal
(D) 4.3 K cal

Q6. Synthesis of ATP in the presence of Oxygen is called :

(A) Cyclic Phosphorylation
(B) Non-cyclic Phosphorylation
(C) Reductive Phosphorylation
(D) Oxidative Phosphorylation

Q7. The Phylum in which animals are exclusively marine :

(A) Cnidaria
(B) Poriphera
(C) Echinodermata
(D) Annelida

Q8. The worm that damages wood of ships is called :

(A) Hookworm
(B) Sepia
(C) Mytilus
(D) Teredo

Q9. Technically a seed may be defined as a fertilized :

(A) Egg
(B) Ovule
(C) Ovary
(D) Flower

Q10. Carcinogenic mycotoxins called aflatoxins are produced by :

(A) Aspergillus
(B) Penicillium
(C) Neurospora
(D) Ustilago

Q11. In 1861, John Hogg proposed the Kingdom :

(A) Protista
(B) Protoctista
(C) Monera
(D) Fungi

Q12. Some Bacteria transfer genetic material from donor to recipient Bacteria during a process :

(A) Binary Fission
(B) Budding
(C) Regeneration
(D) Conjugation

Q13. The Enzyme involved in viral replication is synthesized :

(A) On Viral Capsid
(B) By the Host Cell
(C) On the interior side of Viral coat
(D) On the interior of Viral membrane

Q14. The Undifferentiated cells (such as eggs) have numerous pores about ____ per Nucleus :

(A) 300
(B) 30,000
(C) 6 or 8
(D) 3 or 4

Q15. The type of energy lowered by enzymes for Biological reactions to occur is called :

(A) Kinetic
(B) Potential
(C) Ionic
(D) Activation

Q16. Which of following Fatty Acid is Unsaturated :

(A) Acetic Acid
(B) Oleic Acid
(C) Butyric Acid
(D) Palmitic Acid

Q17. Populations of different Species (Plants & animals) Living in the same Habitat form a :

(A) Community
(B) Tribe
(C) Committee
(D) Population

Biology (Subjective) — Inter (Ist – A – Exam – 2024)

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Note: It is compulsory to attempt any (8 – 8) Parts each from Q.No. 2, Q.No.3 and attempt any (6) Parts from Q.No.4. Attempt any (3) Questions from Part – II. Write same Question No. and its Part No. as given in the Question Paper.

Note: Part - I — 22 x 2 = 44 — Make Diagram where necessary.

Q2.

(i) What is Heat of Vaporization? Give its role in plants.

(ii) What is the difference between Enzyme and Substrate?

(iii) Differentiate between the Reversible and Irreversible Inhibitors.

(iv) What do you know about Lock and Key Model of Enzyme Action?

(v) "Fungi are active predator". Justify this statement.

(vi) Give economic importance of yeast.

(vii) Write down two differences between Diploblastic and Triploblastic animals.

(viii) Give any four Characteristics / features of Class Amphibia.

(ix) What is Haemocyanin?

(x) How insects are important to mankind?

(xi) How would you differentiate between the Aerobic and An-Aerobic Respiration?

(xii) What are Cytochromes?

Q3.

(i) Differentiate between the Deductive Reasoning and Inductive Reasoning.

(ii) Define Biological Control. Give example.

(iii) What are Microtubules? Give their chemical composition.

(iv) Define Chromoplasts. Write their function.

(v) What is Pellicle? Give its functions.

(vi) How Algae are classified? Write names of their Phyla.

(vii) Why Phytophthora infestans is famous for?

(viii) Write Pigments of Red Algae.

(ix) Define Pleura. Give its function.

(x) What are Parabronchi? Write their functions.

(xi) Differentiate between the terms Source and Sink in Plants

(xii) What are Granulocytes? Give their types.

Q4.

(i) Differentiate between Virion and Prion.

(ii) What are Mesosomes? Give its two functions.

(iii) What is Alternation of Generation? Also mention its importance for Plants.

(iv) Define Double Fertilization. Give its Evolutionary Importance.

(v) Write the Scientific name of Lemon Grass.

(vi) Evolution of Pollen tube is great success of Plants on land. Justify.

(vii) What is Peristalsis? How it is different from Antiperistalsis?

(viii) Differentiate between Digestion and Assimilation.

(ix) Teeth are related to feeding habit in animals. Justify.

> Note: Part - II — (3 x 8 = 24)

Q5.

(a) [4] How Biological Science is applied in the field of Food Production?

(b) [4] What is Photorespiration? Give its consequences.

Q6.

(a) [4] Define Proteins. Describe Primary and Secondary Structures of Protein.

(b) [4] What are Imperfect Fungi? Discuss reproduction in Penicillium.

Q7.

(a) [4] Describe Structure and Functions of Mitochondria.

(b) [4] Explain role of Pancreas and Gastric Gland in Digestion of Food.

Q8.

(a) [4] What is Binomial Nomenclature? Who devised this system? Give its rules.

(b) [4] How soil water reaches Xylem tissues by various pathways? Explain.

Q9.

(a) [4] Explain major Characteristics of Cyanobacteria.

(b) [4] How ATP is generated by Cyclic Phosphorylation? Sketch its Diagram.